

---

# Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by  
2 measuring the convergence of foundation-model representations on galaxy images  
3 using Mean-KNN (MKNN) and Centered Kernel Analysis (CKA). We ask whether  
4 that convergence is concentrated where the embedding manifold is flat. A point-  
5 wise curvature field, obtained by differentiating a trained decoder and validated  
6 against analytic truth in that regime, finds mean-curvature magnitude correlating  
7 negatively with cross-survey neighbour agreement at every latent dimension, with  
8 local density explaining 50 to 80 percent of it. A second-order expansion of a  
9 linear map restricted to the manifold explains why: a readout sees the second fun-  
10 damental form contracted with its normal component,  $\langle w_N, \Pi \rangle$ , and an alignment  
11 sees the relative second fundamental form of the two embeddings, neither visible  
12 to mean-curvature magnitude at large codimension. On a unit-normalized embed-  
13 ding that Hessian splits into a sphere term, proportional to the readout’s own value,  
14 and a shape term. On a synthetic surface at production scale the sphere term reads  
15  $-0.74$  to  $-0.80$  across samplings that swing the  $\|H\|$  partial from  $-0.29$  to  $+0.26$ ;  
16 on 86,471 labelled galaxies it is negative on every label, the mismatch between the  
17 label’s own Hessian and the shape term is negative wherever the label has local  
18 structure, and the shape term alone takes the label-dependent sign that mismatch  
19 predicts. Mean curvature of a single representation is therefore not a viable expla-  
20 nation for weak Platonic convergence; the geometric obstruction, where it exists,  
21 is relative and begins at first order.

## 22 1 Introduction

23 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data  
24 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:  
25 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by  
26 a family of different vision models share 0.4 to 2 percent of their nearest neighbours. We ask where  
27 on the manifold that agreement lives: if two models disagree most where the embedding surface  
28 bends sharply, local mean curvature should associate negatively with local cross-survey agreement.  
29 We test this claim using a pointwise mean-curvature estimation from a trained autoencoder on the  
30 embedding space.

31 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a  
32 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled  
33 test of the curvature–alignment prediction under two probes, which finds no effect free of density;  
34 (iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map  
35 responds to and shows mean-curvature magnitude is blind to it; and (iv) a known-surface and a real-

36 data test on the linear-readout side, where the expansion’s terms behave as predicted and account for  
 37 a label-dependent sign.

## 38 2 Pointwise curvature from a decoder

39 **Data.** DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both  
 40 HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], from the Platonic Universe re-  
 41 lease [Duraphe et al., 2025], from which we draw 10,000 rows once with a recorded seed. A second  
 42 set of 86,471 single-survey ViT-B embeddings with catalogue labels ( $r$ -band magnitude, photometric  
 43 redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the read-  
 44 out experiment. Embeddings are unit-normalized (data on  $S^{767}$ ). We then estimate the manifold’s  
 45 first homology ( $\beta_1 = 0$  under bootstrap bands [Fasy et al., 2014]), thus we one Euclidean chart is  
 46 appropriate for the autoencoder.

47 **Instrument.** Let  $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$  be the decoder of a plain auto-encoder (three hidden lay-  
 48 ers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With  
 49  $J = DF(z)$ , metric  $g = J^\top J$ , normal projector  $P_N = I - Jg^{-1}J^\top$  and second fundamental form  
 50  $\Pi = P_N D^2 F(z)$ , the mean-curvature vector is  $H = \text{tr}_g \Pi$ , by automatic differentiation checked  
 51 against finite differences. On the unit sphere every  $d$ -submanifold carries a radial component of  $H$   
 52 equal to  $-d$  that says nothing about shape. To account for this, the map differentiated is  $F/\|F\|$   
 53 (after which  $H_{\text{rad}} = -d$  to  $10^{-14}$ ) and  $\|H_{\text{tan}}\|$  carries the shape. Decoder curvature has served as  
 54 a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is an  
 55 instrument, validated against analytic fixtures.

56 **Validation.** Scored on surfaces with closed-form curvature on four axes: rank (Spearman against  
 57 true  $\|H\|$ ), direction (cosine of  $H$  vectors), magnitude ratio and calibration. At  $D = 768$  an in-  
 58 sphere surface  $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$  ( $\phi$  inverse stereographic  $\mathbb{R}^{16} \rightarrow S^{16}$ ,  $h$   
 59 four Gaussian bumps,  $Q$  a rotation; 18-dimensional span) at  $n = 86,471$ ,  $k = 2,048$  gives rank 0.94,  
 60 direction cosine 0.999, magnitude ratio 1.00, and 0.75 rank under noise at 25% of patch radius.

## 61 3 Curvature and cross-survey convergence

62 **Probe.** Per-point mutual  $k$ -nearest-neighbour score  $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$  at  $k =$   
 63 20 between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local,  
 64 correlated with  $\|H\|$  over all 10,000 points at  $d \in \{20, 25, 32\}$ . Centered kernel alignment [Kornblith  
 65 et al., 2019] is the second probe.

66 **Result.** The raw association  $\rho(\|H\|, \text{MKNN})$  is  $-0.112$ ,  $-0.128$  and  $-0.024$  at  $d = 20, 25, 32$   
 67 ( $p < 10^{-5}$ ,  $< 10^{-5}$ , 0.009), in the predicted direction. But  $k$ -NN statistics are density-  
 68 sensitive [Radovanović et al., 2010]: density correlates with agreement at  $-0.21$  at every  $d$  and with  
 69  $\|H\|$  at  $+0.43$ ,  $+0.32$  and  $+0.01$ , so residualizing on density leaves  $-0.024$ ,  $-0.066$  and  $-0.022$ :  
 70 78%, 49% and 8% of the association removed. Under a permutation null within density strata only  
 71  $d = 25$  survives ( $p < 5 \times 10^{-5}$ ; replicating across three decoder seeds; the CKA tertile gap within  
 72 strata agrees at  $d = 20$  and 25,  $-0.05$  to  $-0.08$  against a null band of  $\pm 0.017$ ). Replacing  $\|H\|$  by  
 73  $\|H_{\text{tan}}\|$  collapses the  $d = 25$  partial to  $-0.023$ , into the range of the two  $d$  that fail. Convergence is  
 74 not detectably concentrated by mean-curvature magnitude once density is controlled.

## 75 4 What a linear map sees on a curved manifold

76 Let  $M \subset \mathbb{R}^d$  be  $d$ -dimensional; at  $x_0$  take an orthonormal tangent frame  $J$ , the second fundamental  
 77 form  $\Pi$  (normal-valued, symmetric  $d \times d$ ),  $H = \text{tr } \Pi$ , and for a vector  $w$  its tangent and normal parts  
 78  $w_T + w_N$ . In normal coordinates  $u$  at  $x_0$ ,  $M$  is  $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$ , so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

79 To second order, a linear function is as non-linear on  $M$  as  $M$  bends in that function’s normal direc-  
 80 tion [Lee, 2018]. For  $M \subset S^{D-1}$  the normal bundle splits into the sphere’s radial direction and the

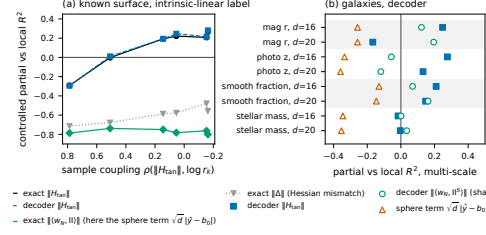


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local  $R^2$  as the sample’s density–curvature coupling is set by construction ( $\gamma$  from  $-1$  to  $1$ ). (b) Galaxies, 512 anchors,  $k = 2,048$ : partial under the multi-scale density control for the decoder’s  $\|H_{\tan}\|$  (filled squares), the shape term (open circles) and the sphere term (open triangles).

rest,  $\Pi = \Pi^S - g \hat{x}$ , so  $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$  (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value  $\hat{y} - b_0$ . For a label  $y$  on  $M$ , a global affine probe  $\hat{y} = w \cdot x + b_0$  and a patch  $u \sim N(0, s^2 I_d)$  in normal coordinates (the  $\frac{1}{2}$  below uses Gaussian fourth moments), with  $\bar{c}$  the patch-mean residual,  $\delta = \nabla_M y - J^\top w$ ,  $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$ ,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings  $F, G$  of the same objects in a common chart  $u$  and a linear map  $A$  with  $AJ_F = J_G$ , the same expansion gives  $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$ , with  $\Gamma$  the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced metrics agree; its normal part is  $\Pi_G - P_N^G A \Pi_F$ .

So the curvature a linear map pays for is  $\langle w_N, \Pi \rangle$  or  $\Pi_G - P_N^G A \Pi_F$ ;  $\|H\|$  bounds only a trace and need not correlate with either at codimension 750; the term carries  $s^4$ , so patch radius, hence density, multiplies it; and when the label has intrinsic curvature of its own,  $\Delta$  carries the residual and the curvature piece can take either sign.

## 5 The probe-facing curvature, on a known surface and on galaxies

The readout side can be tested directly:  $w$  is known and  $\Pi$  comes from the decoder. Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at  $\alpha = 100$  from embedding to label; local  $R^2$  inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local  $R^2$  with Freedman–Lane  $p$ -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, as a stronger control, log radius at  $k \in \{16, 64, 256, 1024, 2048\}$ . Per anchor we compute  $\|H_{\tan}\|$  and the probe-facing Hessian  $\|\langle w_N, \Pi \rangle\|_g$  with  $w_N = w - Jg^{-1}J^\top w$ , split by Eq. (2) into its shape and sphere terms.

**Known surface.** The validation surface is resampled with weight  $\|H_{\tan}\|^\gamma$ , so only where the points sit changes (Figure 1a). The  $\|H_{\tan}\|$  partial follows the sampling from  $-0.29$  at coupling  $+0.80$  through zero to a plateau near  $+0.22$ , and the decoder reproduces it. The probe-facing Hessian  $\|\langle w_N, \Pi \rangle\|$  reads  $-0.74$  to  $-0.80$  at every sampling. On this surface it is the sphere term of Eq. (2): rank  $0.996$ – $1.000$  with  $\sqrt{d}|\hat{y} - b_0|$ , the in-sphere part 20 to 40 times smaller (one in-sphere normal direction) and, where resolvable, following  $\|H_{\tan}\|$ . The fixture thus confirms the expansion’s sphere term: an intrinsically linear label loses local  $R^2$  where the readout’s own value is large, whatever the sampling. With a nonlinear label the sphere term reads null and  $\|\Delta\|$  carries the partial ( $-0.2$  to  $-0.6$ ), as Eq. (3) predicts.

**Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank  $0.96$ – $0.99$  with the full norm). To read it through Eq. (3) we estimate  $\text{Hess}_M y$  per anchor by a least-squares quadratic of the label on the tangent-projected coordinates  $u = g^{-1}J^\top(x - x_0)$  of its 2,048 neighbours, where the quadratic coefficient is the covariant Hessian. Three regularities (Table 1). The sphere term is negative on every label and both  $d$  ( $-0.13$  to

| label           | $\ H_{\text{tan}}\ $ |        | shape $\ \langle w_N, \Pi^S \rangle\ $ |        | sphere $\sqrt{d} \hat{y} - b_0 $ |       | mismatch $\ \Delta\ $ |        | alignment |        |
|-----------------|----------------------|--------|----------------------------------------|--------|----------------------------------|-------|-----------------------|--------|-----------|--------|
|                 | $d=16$               | $d=20$ | 16                                     | 20     | 16                               | 20    | 16                    | 20     | 16        | 20     |
| mag_r           | +0.25                | -0.17  | +0.12                                  | +0.19  | -0.26                            | -0.26 | -0.39                 | -0.40  | +0.35     | +0.43  |
| photo_z         | +0.28                | +0.13  | -0.06*                                 | -0.12  | -0.33                            | -0.36 | -0.45                 | -0.44  | +0.26     | +0.24  |
| smooth_fraction | +0.21                | +0.15  | +0.07*                                 | +0.16  | -0.13                            | -0.15 | -0.09*                | -0.17  | +0.19     | +0.07* |
| stellar_mass    | -0.02*               | -0.00* | +0.00*                                 | +0.03* | -0.35                            | -0.35 | -0.04*                | -0.09* | +0.01*    | +0.02* |

Table 1: Galaxies, 512 anchors,  $k = 2,048$ : rank-partial Spearman against local  $R^2$  under the multi-scale density control; alignment is  $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ . \* not significant at 0.05; the three-control partials agree in sign in every significant cell.

−0.36), as on the fixture. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is null throughout) the mismatch is negative (−0.09 to −0.45) and the alignment  $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$  positive (+0.07 to +0.43): local  $R^2$  is higher where the manifold bends in the readout’s direction the way the label bends. The shape term alone is therefore label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift) because it is one of three terms in  $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$ : its sign follows the cross term’s where that is nonzero and the remaining square where it is null.  $\|H_{\text{tan}}\|$  changes sign between  $d = 16$  and 20 for magnitude: it sits on the curve of Figure 1a (coupling to radius −0.60). The same regression applied to the readout  $w \cdot x$  returns the point cloud’s own  $\langle w_N, \Pi \rangle$  (median metric cosine with the decoder’s 0.55–0.65), and an independent split-half quadratic-chart estimator of  $\Pi$  gives the same shape-term sign on three of four labels.

**Across encoders.** The quadratic-chart estimator’s mean-bending trace, run on five encoders at the same 512 anchors ( $d = 16$ ,  $k = 2,048$ : ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L), gives density-controlled associations with the magnitude probe’s local  $R^2$  of −0.24, 0.00, +0.27, +0.23 and −0.19. A trace statistic has no encoder-stable sign, as Eq. (1) predicts; a cross-encoder test should carry the terms of Eq. (3).

## 6 Discussion

Three things are established. A trained decoder yields a pointwise second fundamental form at production scale whose readout-facing component agrees with an independent estimator and with the point cloud itself. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the sphere term and the Hessian mismatch carry the predicted negative sign, and the shape term is label-signed because it enters through its alignment with the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. For the Platonic question this fixes the design. MKNN and CKA are invariant to rigid motions of either embedding, so a per-point convergence score can be conditioned on curvature only through a relative quantity such as  $\Pi_G - P_N^G A \Pi_F$ , which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot at  $d = 20$  (2,048 anchors) supplies both and locates the obstruction to linear alignment at first order: a global ridge map carries HSC tangent frames into the Legacy embedding with 44% residual, and the second-order term shows no density-free association yet (+0.04, n.s.; −0.18 to −0.24 with a locally fitted tangent map, +0.13 decoder-free). The explicit finding is therefore: mean curvature of a single representation does not explain weak Platonic convergence; the geometric obstruction is relative, begins at first order, and testing it means aligning tangent frames first and then the relative second fundamental form, for which the instrument is in hand.

**Limitations.** One survey pair. The label’s Hessian is a local estimate ( $R^2$  gain over linear 0.10–0.17; a 136-coefficient fit of the same kind failed a split-half reliability gate). The sphere term is not separated from ridge shrinkage of extreme predictions. Permutation  $p$ -values assume anchor exchangeability under overlapping patches.

## References

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://>

158     [//arxiv.org/abs/2405.07987](https://arxiv.org/abs/2405.07987).

159     Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation  
160         models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025.  
161         URL <https://arxiv.org/abs/2509.19453>.

162     Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv*  
163         *preprint*, 2025. URL <https://arxiv.org/abs/2508.10104>.

164     Hiroaki Aihara, Nobuo Arimoto, Robert Armstrong, et al. The Hyper Suprime-Cam SSP Survey: Overview  
165         and survey design. *Publications of the Astronomical Society of Japan*, 70(SP1):S4, 2018. doi: 10.1093/pasj/  
166         psx066.

167     Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI Legacy Imaging Surveys. *The Astro-*  
168         *nomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

169     Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large obser-  
170         vation models for astronomy. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2405.14930>.

171     Brittany Terese Fasy, Fabrizio Lecci, Alessandro Rinaldo, Larry Wasserman, Sivaraman Balakrishnan, and Aarti  
172         Singh. Confidence sets for persistence diagrams. *Annals of Statistics*, 42(6):2301–2339, 2014.

173     Yonghyeon Lee and Frank Chongwoo Park. On explicit curvature regularization in deep generative models. In  
174         *2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, ICML, 2023.  
175         URL <https://arxiv.org/abs/2309.10237>.

176     Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic  
177         curvature in neural manifolds. *arXiv preprint*, 2022. URL <https://arxiv.org/abs/2212.10414>.

178     Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network rep-  
179         resentations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*,  
180         2019.

181     Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors  
182         in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531, 2010.

183     John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer,  
184         2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.

185     Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.  
186         arXiv:1610.01644.

187     David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of*  
188         *Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.